Financial Text Summarization & Information Extraction

Lecture 14 — Turning Filings, Calls, and Reports into Structured, Faithful Facts

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Outline

Why Extraction and Summarization Matter

THE BIGGER PICTURE

Markets run on information, not on its absence. Tens of thousands of filings, transcripts, and analyst notes are released daily. The bottleneck is the *cost of reading, interpreting, and routing* — not the lack of facts.

The scale problem.

- A PM cannot read every 10-K from the 6,000+ firms in the Russell 3000.
- A compliance officer cannot scan every contract in a merger data room.

Three tightly coupled IE sub-tasks:

1. **Entity recognition** — who/what the text refers to.
2. **Fact / relation extraction** — how entities relate, what numbers they carry.
3. **Summarization** — condense long documents, faithfully.

Dependency chain

NER quality *caps* relation-extraction accuracy; faithful summarization *requires* accurate entity and fact extraction underneath.

Structured vs. Semi- vs. Unstructured Text

Definition — the structure spectrum

A document is **structured** if every element can be queried programmatically via a published schema (no NL needed); **semi-structured** if it mixes schema-conformant sections with free-form narrative; **unstructured** if its content lives in natural language with no imposed schema.

Tier	Examples	Strategy
Structured	XBRL filings, tick data, options chains	Database query
Semi-structured	10-K / 10-Q, transcripts, proxies	Tables + NLP
Unstructured	News, social media, analyst notes, contracts	Full NLU

Practitioner rule

Start with structured sources (low noise, low cost). But unstructured text holds *forward-looking* signal — guidance, qualitative risk, interpretation — absent from the tables. The best pipelines combine both.

Key Entities in Financial Text

Organisations — most frequent, most ambiguous. "Apple" (firm vs. fruit), "Morgan" (J.P. Morgan vs. Morgan Stanley). Resolution = *entity linking* to LEI / CUSIP / EDGAR CIK.

Temporal expressions — "next quarter", "FY2024", "within 30 days of the trigger". Must normalise fiscal + calendar terms to dates.

Monetary figures & metrics — $ amounts, %, EPS, EBITDA margin, D/E. Resolve units (M vs. B), parenthesised negatives, and the entity/period each figure describes.

Legal / contractual — covenants in loan agreements and bond indentures that can trigger default; classify and extract threshold values.

Earnings-call excerpt → extracted records

"In Q3 2023, **Microsoft** reported revenue of **$56.5 billion**, up **13%** YoY. CEO **Satya Nadella** noted **Azure** grew **29%**…"

ORG: Microsoft (CIK 789019) · PERSON: Nadella/CEO · DATE: Q3'23 → 2023-07-01..09-30 · MONEY: $56.5B (total rev) · METRIC: +13% rev, +29% Azure.

Extraction vs. Summarization: A Precise Taxonomy

Information Extraction (IE)

Schema-driven. Identify and structure specified information into a **predefined schema**. Output = typed records (entities, attributes, relations). Models are **discriminative**: classify whether a span fills a slot.

"What happened to whom, when, by how much?" → populates databases.

Summarization

Content-driven. Produce a shorter coherent text capturing the salient content. Output = a **natural-language string**. Models are **generative**; "salient" is learned from reference summaries.

"What should I know without reading it all?" → serves readers.

Why the distinction matters

The two have *different evaluation criteria and failure modes*. A quant trading earnings surprises needs IE; an investment committee reviewing a note needs summarization; a covenant monitor needs IE *then* a human-readable summary.

Financial NER: Beyond the Four CoNLL Classes

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NER is the foundational IE task: label each token with its entity type (or O). The general-domain CoNLL-2003 schema — PERSON, LOCATION, ORGANISATION, MISC — is *inadequate* for finance.

Financial documents add specialised types: **financial instruments**, **economic indicators**, **legal clauses**, **corporate roles**. These demand domain-specific datasets and fine-tuned models.

Three finance-specific annotation challenges

1. **Context-dependent type**: "Apple" = org (portfolio) vs. product (supply chain).
2. **Nested entities**: "Goldman Sachs Investment Banking Division" = org + unit.
3. **Implicit entities**: "the issuer" refers to a firm named earlier.

Financial NER & RE Datasets

Dataset	Type	Scale / scope	Use case
FINER-139	NER, 139 types	SEC filings + news	Fine-grained NER
FinNER	NER, doc-level	10-K / 10-Q	Coreference chains
FinRE	Relations	(e_1, rel, e_2) triples	Ownership graphs
EDGAR-Corpus	Pretraining	6M+ filings, $\sim$6.5B tok	Domain pretraining

- **FINER-139**: hierarchical taxonomy (Org $\rightarrow$ Public Company, ETF, Interest Rate. . .). "Company" is *not* monolithic — conflating PE firms with listed banks creates systematic graph errors.
- **FinRE** relations: `Subsidiary-Of`, `Acquired-By`, `Founded-By`, `Announced-Transaction`.
- **EDGAR-Corpus**: 10-K/10-Q from 1993–2020. Models pretrained on it *systematically* beat general-domain models on financial NER.

Fine-Tuning a Transformer NER Head

UNDER THE HOOD

Token classification on a pretrained encoder. For tokens $\mathbf{x} = (x_1, \ldots, x_T)$ with contextual states $\mathbf{h}_i \in \mathbb{R}^d$:

$$P(y_i \mid \mathbf{x}) = \mathrm{softmax}(\mathbf{W}\mathbf{h}_i + \mathbf{b}), \quad \mathbf{W} \in \mathbb{R}^{C \times d}.$$

Labels follow **BIO**: $\{B\text{-}e, I\text{-}e, O\}$. Train end-to-end on token cross-entropy $\mathcal{L} = -\sum_t \log p(\hat{y}_t = y_t)$.

Encoder choice matters. FinBERT > vanilla BERT; **SEC-BERT** (pretrained only on EDGAR) is SOTA on SEC-domain NER — it has seen "Regulation S-K Item 1A" thousands of times.

Three finance-specific recipe tweaks beyond the standard:

1. **Segmentation**: docs exceed 512 tokens → sliding windows (≥64-token overlap) or Longformer.
2. **Label imbalance**: O is 85–95% of tokens → focal / weighted loss.
3. **Post-processing**: a CRF + Viterbi layer enforces valid BIO transitions.

Example: SEC-BERT → FINER-139

Production-grade recipe

1. Download FINER-139 splits ($\sim$900 annotated training sentences).
2. Init from SEC-BERT checkpoint (MLM-pretrained on millions of SEC sentences).
3. Linear head of dimension $C = 2 \times 139 + 1 = 279$ (BIO for 139 types + O).
4. AdamW (β_1=0.9, β_2=0.999, wd 0.01), linear warmup 10%, peak LR 2×10^{-5}, 10 epochs.

Result: $\sim$20 min on one A100; macro-F_1 *substantially* exceeds the FinBERT baseline on rare types ($<$50 training examples) — where domain pretraining pays off most.

Relation Extraction: Entities → Facts

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Entities acquire financial meaning through relationships: Microsoft *acquired* Activision; Buffett *is CEO of* Berkshire; a loan is *collateralised by* specific assets. These feed knowledge graphs, earnings attribution, and corporate-hierarchy databases.

UNDER THE HOOD

Joint entity-and-relation models (e.g. DyGIE++) score ordered span pairs:

$$s(e_i, r, e_j) = f\left(\mathbf{h}_{e_i}, \mathbf{h}_{e_j}, \mathbf{r}\right),$$

with span reps $\mathbf{h}_e$ (mean/endpoint of encoder states), learned relation embedding $\mathbf{r} \in \mathbb{R}^{d_r}$, and a bilinear/FFN scorer f; predict $r \in \mathcal{R} \cup \{\text{None}\}$.

Taxonomy: SUBSIDIARY-OF, ACQUIRED-BY, EMPLOYED-AT, BOARD-MEMBER-OF, ISSUED-BY, SECURED-BY, REPORTED-VALUE.

Hard case: *cross-sentence* RE $\gg$ within-sentence. Document-level models (DocRED-style) that encode full context win.

Extractive vs. Abstractive — The Faithfulness Tradeoff

Extractive

Select a subset of sentences/spans **verbatim** from the source. Faithfulness **guaranteed by construction** — every sentence was written by the original author.

Abstractive

Generate **new text** that paraphrases and synthesises. More concise and fluent — but susceptible to **hallucination**: plausible-sounding but factually wrong statements.

Why hallucination is consequential in finance

Summarising "revenue of \$57B" when the truth was \$56.5B is small in % but can be *material misinformation* in a decision process. SOTA abstractive models hallucinate factual errors in **25–30%** of CNN/DailyMail summaries; untuned financial models fare no better.

Recommendation: prefer extractive when faithfulness is paramount (esp. numbers); use abstractive when fluency dominates. **Hybrid** (extract, then rephrase) is a useful middle ground.

Earnings & 10-K Summarization

A typical 10-K runs **50,000–150,000 words**: business, risk factors, legal proceedings, MD&A, financial statements, notes.

The MD&A is the densest section for an equity investor — management's interpretation and forward guidance. Its tone, readability, and length carry measurable signal about future earnings and returns (Loughran–McDonald). Tone-preservation must be an *explicit* constraint in any faithful prompt.

Three-stage pipeline

1. **Section extraction & routing** — parse TOC, route each section to a sub-model (table extractor vs. text summariser).
2. **Chunk-level summarization** — 512-tok chunks (64 overlap) for BERT; whole MD&A in one pass for 8K–128K decoders.
3. **Hierarchical aggregation** — second-pass or retrieval-based merge of chunk summaries into a document summary.

Example: Constrained Prompt for Earnings

Numerical-hallucination-resistant prompt

```
You are a senior equity analyst. Summarise the following earnings
release in no more than 150 words, covering: (1) revenue and EPS
vs. consensus, (2) the single most important forward guidance
statement, and (3) one key risk mentioned by management.
Quote all numbers exactly as stated.

[EARNINGS RELEASE TEXT]
```

The key line: "Quote all numbers exactly as stated" is a practical prompt-engineering technique that reduces numerical hallucination.

Illustrative finding (figures vary with corpus/model): the constrained prompt cut number errors by $\sim$40% vs. an unconstrained paraphrase prompt; $\sim$8% of cases still showed paraphrase errors on nuanced guidance — motivating the *factual-consistency* checks later.

Analyst Notes & Long-Document Challenges

Analyst-note condensation. Reports run 10–40 pp with a predictable macro-structure (thesis in para 1, target price standardised). Template-aware parsing exploits this.

Change detection / delta summarization: a PM wants *what changed*, not what they already know — requires report history + reasoning over diffs.

Earnings transcripts: 15,000–30,000 words; uneven density (prepared remarks dense; Q&A mixes signal + boilerplate). **ECTSum**: 2,425 transcript–summary pairs, 3–6 abstractive bullets each.

S-1 filings: a tech S-1 can exceed **300,000 words**.

Common thread: no single context window fits these documents, so the architecture — not just the prompt — must handle length. Three strategies follow on the next slide.

Long-Document Strategies

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Three strategies for extreme-length documents:

- **Two-stage hierarchical**: each section → 200–400 words, then a second-pass model over the concatenated section summaries.
- **Retrieval-augmented summarization**: dense-retrieve top-k passages for a query, summarise only those (summarization as RAG).
- **Long-context models**: Longformer; 128K–1M-token models (e.g. Gemini 1.5 Pro) process an entire S-1 in one pass — cost-bound.

SEC XBRL: Structured Extraction Without NLP

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Tables are 2-D grids where each cell's meaning depends on its row header (what measure) *and* column header (what period/entity). Reliable table extraction underpins database population and consistency checking.

Inline XBRL (mandated for U.S. domestic filers since 2020): every figure tagged with a U.S.-GAAP concept (e.g. `us-gaap:Revenues`), a context (entity + period), and units — machine-readable *and* browser-readable.

Prefer XBRL whenever available

The EDGAR XBRL API returns a figure with full provenance for a (CIK, concept, year) query. Accuracy is guaranteed by the filer and its auditors — *no NLP*.

Coverage gap

Non-GAAP measures (adjusted EBITDA, organic growth, book-to-bill) and many footnotes are *not* XBRL-tagged. Combine XBRL for core figures with LLM/NLP extraction for the surrounding narrative.

LLMs for Table Understanding

When XBRL is unavailable (international filings, private docs, pre-XBRL legacy): tables must be parsed from HTML/PDF/text — merged cells, multi-row headers, embedded footnote refs, non-standard layouts.

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FinQA benchmarks multi-step numerical reasoning over tables:

- Baseline cell-retrieval models: $\sim$50% exact-match.
- Fine-tuned models that *generate a Python-like calc program*: $>$70%.

Why generation beats retrieval: financial table questions chain several operations (subtract prior year, divide, express as %). A model that emits an explicit calculation program externalises that reasoning and routes the arithmetic to a deterministic interpreter — continued next slide.

LLMs for Table Understanding — Pipeline

Four-stage LLM pipeline for parsing tables when XBRL is unavailable:

1. **Layout parsing** (LayoutLM / PDF parser) $\rightarrow$ cells with row/col coords.
2. **Header disambiguation** $\rightarrow$ LLM labels each cell with (row-hdr, col-hdr).
3. **Value normalisation** $\rightarrow$ units, parenthesised negatives, % vs. absolute.
4. **Schema mapping** $\rightarrow$ map to GAAP taxonomy concepts.

Arithmetic reliability (ReAct principle)

LLMs are unreliable calculators. **Route all arithmetic to a code interpreter**: the LLM decides *what* to compute; Python *actually* computes it. This separation dramatically cuts numerical errors.

Consistency Checking: Numbers vs. Prose

The same revenue figure appears in the tables, the press release, and the MD&A. **Consistency checking** verifies they agree — catching (a) *extraction errors* and (b) *genuine disclosure inconsistencies* (possibly material).

Three-step procedure

1. **Multi-source extraction** — pull every occurrence of each metric (revenue, op. income, EPS) from tables, press release, MD&A, footnotes.
2. **Normalisation** — common unit & sign. "\$56.5B" → \$56,500M; all sources should agree.
3. **Cross-reference** — flag any pair disagreeing beyond tolerance (e.g. 0.1% for rounding); report with source attribution.

Prose vs. table fact-checking. "Net income grew 25% YoY" can be verified against extracted current/prior figures — a powerful quality gate for both document pipelines *and* LLM-generated summaries, where hallucinated numbers are a known failure mode.

ROUGE — Recall-Oriented Overlap

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ROUGE is the most widely used automatic summarization metric. Intuitively: how much of the *reference* summary's wording does the candidate reproduce? A recall-oriented overlap score.

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ROUGE-N = n-gram recall of candidate c against references $\mathcal{R}$ ($\mathcal{G}_N(s)$ = n-grams of s):

$$\text{ROUGE-}N = \frac{\sum_{s\in\mathcal{R}} \sum_{g\in\mathcal{G}_N(s)} \text{Count}_{\text{match}}(g, c)}{\sum_{s\in\mathcal{R}} \sum_{g\in\mathcal{G}_N(s)} \text{Count}(g)}.$$

ROUGE-L uses the longest common subsequence instead of exact n-grams.

ROUGE — Why Lexical Overlap Is Not Enough

ROUGE rewards *surface n*-gram overlap with the reference. That makes it cheap and reproducible, but blind to meaning.

Limitation: lexical, not semantic

"net income declined 12%" $\rightarrow$ "profit fell 12%" deserves full credit but scores **zero** ROUGE-1 on the substituted words. Conversely, copying irrelevant source sentences can score *high* ROUGE without being informative.

Takeaway: treat ROUGE as a coarse recall proxy, not a quality verdict — it motivates the semantic and factual metrics that follow.

BERTScore — Semantic Matching

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BERTScore fixes ROUGE's lexical-overlap blind spot: candidate and reference tokens count as equivalent when their *meanings* match, not only their surface forms.

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Cosine similarity of contextual embeddings for candidate token $\hat{r}_j$ and reference token r_i:

$$\mathrm{sim}(r_i, \hat{r}_j) = \frac{\mathbf{h}_{r_i}^{\top} \mathbf{h}_{\hat{r}_j}}{\|\mathbf{h}_{r_i}\| \, \|\mathbf{h}_{\hat{r}_j}\|}.$$

Precision/recall/F_1 via greedy token matching in both directions. With a *financial* encoder (FinBERT), it captures domain-specific similarity better.

BERTScore — What Semantic Matching Still Misses

BERTScore closes ROUGE's paraphrase gap, but similarity in embedding space is still not the same as being *factually correct*.

Neither ROUGE nor BERTScore measures factual correctness

A summary inventing "revenue grew 15%" (true: 13%) can still score high if the surrounding language matches the reference. And outputs can be inconsistent *across runs* — reproducibility itself is a QA concern for production IE.

Takeaway: similarity metrics rank fluency and coverage, not truth — hence the factual-consistency checks on the next slide.

Factual Consistency & Human Evaluation

Factual-consistency metrics

FactCC trains an NLI classifier (entailment / contradiction / neutral) to predict whether each candidate claim is *entailed* by the source. For finance, strengthen by **grounding numerical claims against extracted XBRL**: any claim that cannot be verified against structured data is flagged.

Human evaluation — the gold standard — by *domain professionals* (a crowdworker cannot judge GAAP vs. non-GAAP revenue). Assess four dimensions:

- **Factual accuracy** (most critical; mark each claim verified / unverifiable / incorrect)
- **Completeness** — all material disclosures covered?
- **Conciseness** — no redundancy?
- **Fluency / readability**

Report agreement (Cohen's κ / Krippendorff's α): typically *low* on completeness (analysts value different facts), *high* on factual accuracy (errors are objective) $\Rightarrow$ accuracy is the tractable auto target.

Inventory of Financial NLP Datasets

Dataset	Document type	Scale	Primary task
ECTSum	Earnings transcripts	2,425 pairs ($\sim$7k words ea.)	Long-doc summarizatior
EDGAR-Corpus	SEC filings	6M+ filings	Domain pretraining
FINER-139	SEC filings + news	139 entity types	Fine-grained NER
FinSBD	Financial text	annotated	Sentence boundary detec
FinQA	Tables + text	multi-step Q&A	Numerical reasoning

No single benchmark covers the domain

A model that tops ECTSum (summarization) may fail on FinQA (table reasoning). Read "SOTA on financial NLP" *with the specific benchmark in mind*. Match the dataset to your document type and task — and construct a *held-out* test set from documents outside any public benchmark before deploying.

Lecture 14 — Five Takeaways

1. **Financial text is heterogeneous** — structured (query) to unstructured (full NLU). Production pipelines combine both, anchored on structured sources.
2. **Financial NER needs domain data & models** — general-domain NER is inadequate for the 139-class taxonomy. Pretrain on EDGAR, fine-tune on FINER-139.
3. **Abstractive summarization carries hallucination risk** — in finance, hallucinated numbers can be *misinformation*. Pair with factual-consistency checking against structured data.
4. **Tables & numerical consistency are underappreciated** — XBRL is free and accurate; LLMs supplement for non-GAAP & legacy; route arithmetic to code.
5. **Match the metric to the stakes** — ROUGE/BERTScore measure similarity, not facts. Add factual-consistency checks *and* expert human evaluation.

Next

Practical 14: an end-to-end SEC 10-K pipeline — section routing, extractive summarization, ROUGE scoring, and consistency checking.

Appendix

Extended Material — Annotation, RE Internals, Further Reading

Extra / optional material — beyond the core lecture.

Appendix: Annotation Challenges in Finance

General-domain NER guidelines do not anticipate these:

- **Surface-string ambiguity**: "Apple" as organisation (portfolio) vs. product (supply-chain discussion) — the *same string*, different type.
- **Nested entities**: "Goldman Sachs Investment Banking Division" = org (Goldman Sachs) + business unit, simultaneously.
- **Implicit entities**: "the issuer" in a bond prospectus refers to a previously named company without restating it.

These motivate *domain-specific annotation guidelines and adjudication processes* — not just relabelled CoNLL guidelines.

Appendix: Why Cross-Sentence RE Is Hard

- Within-sentence RE: both entities co-occur $\rightarrow$ local context suffices.
- Cross-sentence RE: entities in different sentences $\rightarrow$ requires *document-level* encoding and coreference resolution.
- Benchmarks: DocRED and its financial adaptations; document-level models that encode the whole context (not sentence-by-sentence) achieve the gains.

Why it matters in finance: ownership chains and "the issuer"-style implicit references routinely span paragraphs in prospectuses and 10-Ks.

Appendix: Retrieval-Augmented Summarization

For documents that exceed any context window (S-1 >300k words), turn summarization into **retrieval-augmented generation**:

1. Index the document's passages with a dense retriever.
2. For a query ("What are the principal risk factors?"), retrieve top-k.
3. Summarise *only* the retrieved passages.

Tradeoff vs. hierarchical summarization: RAG is query-targeted and cheap but may miss globally salient content not matching the query; hierarchical two-stage covers everything but costs a full pass over the document.

Appendix: Further Reading

- **NER foundations**: Lample et al. (2016); Devlin et al. (2019) survey. Financial: FINER-139 (Loukas 2022); English/Greek FINER (Shah 2023).
- **Summarization**: ECTSum (Mukherjee 2022) for transcripts; EDGAR-Corpus (Loukas 2021) for domain-adaptive pretraining.
- **Evaluation**: ROUGE (Lin 2004); BERTScore (Zhang 2020); FactCC (Kryscinski 2020); faithfulness analysis (Maynez 2020).
- **Table reasoning**: FinQA (Chen 2021); broad survey FinBen (Xie 2024).

Cross-chapter links: XBRL + table parsing here are prerequisites for the DCF pipeline in the Business Valuation chapter; apply the hallucination-mitigation framework (LLM Limitations chapter) to any decision-critical abstractive summary.